

# David Bean

## DATA SCIENCE, DATA PIPELINES

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### SKILLS

|                         |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| <b>Languages</b>        | Python, SQL, Bash, R                                                                                 |
| <b>Acquisition</b>      | REST APIs, web scraping, JSON, requests, BeautifulSoup, web crawlers, CSV, geospatial data, NetCDF   |
| <b>Wrangling</b>        | pandas, NumPy, validation suites, merging sources, deduplication, SciPy, GeoPandas, dplyr, xarray    |
| <b>Pipelines</b>        | incremental processing, parallelization, larger-than-memory processing (100+ GB), checkpointing      |
| <b>Linux</b>            | parallel batch jobs, SSH, Slurm, grep, sed, awk, Make, tmux                                          |
| <b>Machine Learning</b> | scikit-learn, XGBoost, feature selection, cross-validation, gradient boosting, PCA, k-means, UMAP    |
| <b>Statistics</b>       | hypothesis testing, linear regression, logistic regression, experimental design, t-tests, chi-square |
| <b>Visualization</b>    | interactive dashboards, matplotlib, Google Sheets, Excel, Jupyter, ggplot2, Vega-Altair, RStudio     |
| <b>Databases</b>        | schema design, joins, DDL, ER diagrams, GROUP BY, primary and foreign keys                           |
| <b>Version Control</b>  | Git, GitHub, pull requests, GitLab, branching and merging                                            |
| <b>Containers</b>       | Docker, Dockerfiles, GPU containers, Podman, Apptainer, venv                                         |

### EDUCATION

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|-------------------------------------------------------------------------------------|----------|
| <b>Master of Science in Biomedical Informatics</b>                                  | May 2026 |
| University of Utah   Salt Lake City, UT   Emphasis: Data Science and AI   GPA: 3.95 |          |
| <b>Bachelor of Science in Biology, magna cum laude</b>                              | May 2023 |
| Utah Tech University   Saint George, UT   Emphasis: Biomedical Sciences   GPA: 3.80 |          |

### EXPERIENCE

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| <b>Research Assistant II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jun 2025 - Dec 2025 |
| University of Utah - Department of Biomedical Informatics, Salt Lake City, UT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |
| <ul style="list-style-type: none"><li>• Diagnosed a published mutation classifier's precision drop from 0.71 to 0.02 on patient data as sensitivity to class ratio</li><li>• Optimized pipeline runtime 10x (~50 to 5 hours) via parallel, containerized Slurm jobs and a resumable task scheduler</li><li>• Eliminated ~90% of harmless mutations per patient via population databases, retaining 96% of known cancer mutations</li><li>• Defended investment in the research to department leadership in a 30-minute talk distilling 6 months of work</li></ul>                             |                     |
| <b>Research Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Oct 2023 - Apr 2024 |
| Utah Tech University - Klabacka Lab, Saint George, UT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |
| <ul style="list-style-type: none"><li>• Automated retrieval of protein data for 4,747 genes across 5 species by REST API in a Dockerized Bash and Python pipeline</li><li>• Batch-generated data visualizations in R (ggplot2), looping one reusable chart design over data from 141 bacterial species</li><li>• Demonstrated a vital protein ranked below 56% of 2,257 genes in genetic similarity, highlighting key positions for survival</li><li>• Unified code from multiple contributors into one automated pipeline and presented findings at 3 research poster sessions</li></ul>     |                     |
| <b>Laboratory Instructor, Biology 1625</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Aug 2023 - May 2024 |
| Utah Tech University - Department of Biological Sciences, Saint George, UT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |
| <ul style="list-style-type: none"><li>• Led 3 lab sections of 18 to 24 students through full research cycles, from posing a question to presenting for peer feedback</li><li>• Taught students to state null and alternative hypotheses, design experiments, and draw conclusions from statistical tests</li><li>• Lectured on choosing the right statistical test for a dataset (paired and unpaired t-test, chi-square, linear regression)</li><li>• Trained students to combine statistical tests with multiple visualizations to reveal patterns from more than one perspective</li></ul> |                     |

### PROJECTS

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| <b>Interactive Monarch Migration Map</b>                                                                                                                                                                       | 2024 |
| <ul style="list-style-type: none"><li>• Told the story of monarch migration in an interactive Vega-Altair dashboard of sightings and weekly temperature maps</li><li>• Harmonized 227,000+ monarch sightings with the temperature grids by reprojecting both to the Mercator projection</li></ul> |      |
| <b>Motorcycle Crash Risk Analysis</b>                                                                                                                                                                          | 2025 |
| <ul style="list-style-type: none"><li>• Backed all 14 group comparison charts with hard numbers, a statistical test and p value behind each (Cliff's delta 0.75)</li><li>• Merged 7 years of federal crash files on a composite key into 870,000+ person-level records with pandas</li></ul>      |      |